
CE 611: Structural Health Monitoring

Course Code	CE 611
Course Name	Structural Health Monitoring
L-T-P-C	3-0-0-3
Prerequisites	Strength of Materials and Structures – CE 301 and Structural Dynamics with Application to Earthquake Engineering – CE 511 or equivalent
Intended for	UG and PG
Distribution	Discipline Elective (UG and PG)
Approval	16th Senate (Apr 2018), in the M.Tech Structural Engineering curriculum; 28th BoA (Jan 2019); 19th Senate (Feb 2019)

Course Contents

- **Module I:** Introduction to SHM, brief history structural rehabilitation, Condition monitoring, Infrastructure management, Components of SHM, Sensors and Instrumentation, Measurement noise and precision. [4 Lectures]
- **Module II:** Non-destructive evaluation based methods: Acoustic emission based, Ultrasonic testing, Eddy current based method, Radiographic imaging based. [4 Lectures]
- **Module III:** Vibration based methods, basic procedures (Operational evaluation, data acquisition fusion and cleansing), feature extraction, feature discrimination, Modal information based approaches: Model independent methods, frequency based (forward and inverse), Mode shape and modal curvature based methods, Modal information based approaches: Model dependent methods, Direct matrix update, Parameter optimization, soil computing based methods. [15 Lectures]
- **Module IV:** [4 Lectures]